

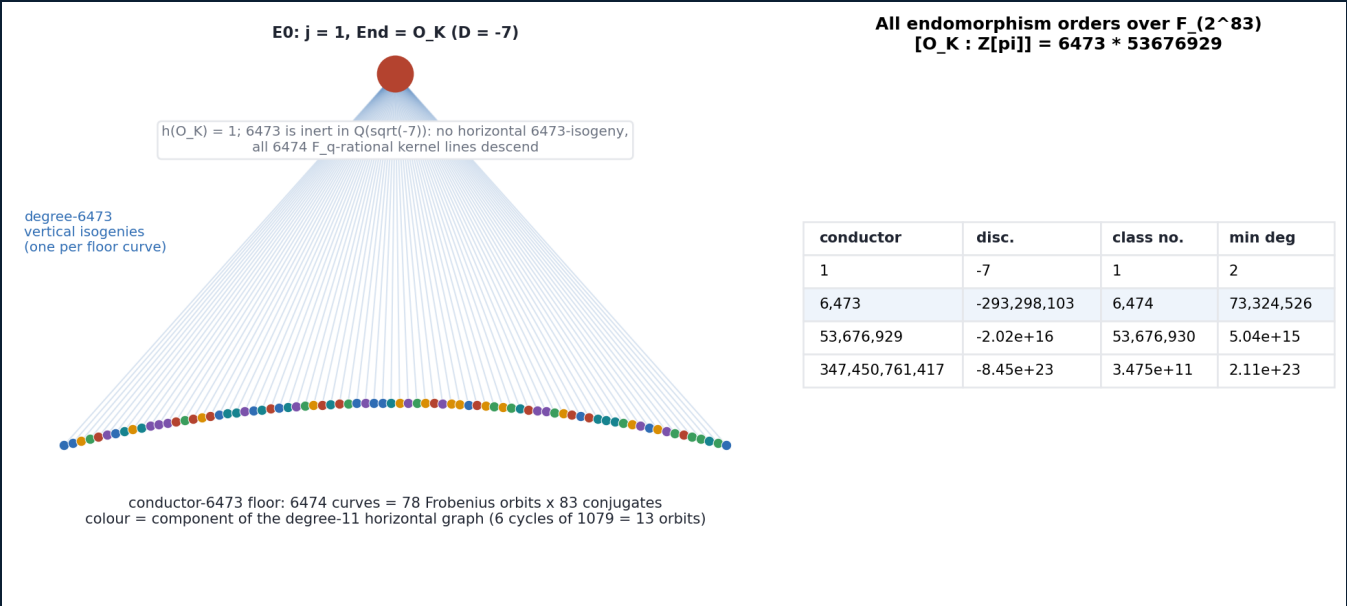
Endomorphism Orders, Factor Bases, Groebner Behavior, and End-to-End Attack Cost

The ECC2K-130 volcano-descendant study repeated for the repository's $m=83$ Koblitz test curve: E_0 over $F_{(2^83)}$, all 6474 conductor-6473 descendants, explicit degree-6473 isogenies, and the vertical, Frobenius-conjugate and horizontal-isogeny mechanisms.

Scope
Public curve arithmetic and reproducible finite experiments.

Status
No discrete logarithm or sensitive key was computed.

Evidence
Runs 01-10: complete-floor inventory, explicit degree-6473 maps, exact relation images, solver traces, atlases and audits.



Volcano and endomorphism-order structure.

Executive assessment

Security conclusion

Descending from the maximal order to the conductor-6473 floor of the F₂⁸³ Koblitz curve changes the geometric endomorphism ring, the local 6473-torsion, the model field of definition and the available self-maps. As for ECC2K-130, those changes produce many finite coordinate and factor-base differences, but no tested descendant mechanism reduces complete index-calculus work. The apparent gains disappear under matched cardinality, adjacent-dimension scaling, matrix-column accounting, or map and solver cost.

The original curve E0 is the Koblitz curve $y^2 + xy = x^3 + 1$ over $F_2(2^{83})$, the m=83 instance of this repository's ECC2K-130 client, with prime subgroup order $ELL = 2,417,851,639,230,796,216,685,689$ (81 bits). Its geometric endomorphism ring is the maximal order of $Q(\sqrt{-7})$. The Frobenius order has index $347,450,761,417 = 6473 \times 53676929$ in O_K , and both primes are inert. The nearest strict descendants have conductor 6473, discriminant -293,298,103 and class number 6,474. All 6,474 floor models were constructed from a ring class polynomial, certified, and matched one-to-one against explicit degree-6473 isogenies from E0.

Main findings

- The prime subgroup and its 74-bit embedding degree are invariant across the isogeny class; descent creates no MOV shortcut.
- Strict descendants lose E0's degree-2 Frobenius self-endomorphism. Their minimum noninteger endomorphism degree is 73,324,526, against 2 on E0 (121,046 at ECC2K-130).
- Because 6473 is inert, E0 has no horizontal 6473-isogeny: all 6,474 rational kernel lines descend, one to each floor curve. The floor is 78 Frobenius orbits of 83 conjugates (ECC2K-130: two orbits of 131).
- Frobenius acts on E0[6473] as the scalar 2514, a primitive root mod 6473, so the kernel x-coordinates live in $F_2(2^{268588})$. Explicit torsion points, all 6474 Velu codomains and degree-3236 kernel polynomials were computed there with native PMULL arithmetic; the codomains reproduce the class-polynomial inventory exactly.
- Exact relation images on all 32,375 curve/profile rows obey the four-torsion eligibility formula; the only non-injective images are E0's, all generated by the single Frobenius identity $2P_w + P_w^2 - P_w^4 = O$ and adding no factor-base rank.
- Formal regularity follows the membership-indicator presentation, not the conductor: at k=5 the maximum sampled degree equals 6 + membership-ANF degree for 160 of 160 panel descendants, while E0 (membership degree 5) sits at 9. Swapping only E0's coefficient b = 1 for floor coefficients restores the stratum, so the anomaly is a crater property.
- Adapted descendant subspaces reach 25%-29% single-cell gains at k=8-9 and two curves out of 6474 clear the adjacent-dimension gate at k=8/9, but every lead decays; by k=16 the best curve is at 0.9772 and the 5%-95% band is 0.9958-1.0340 of optimized E0.
- Frobenius-conjugate atlases (all 78 orbits) and the degree-11 horizontal atlas (a 1079-chart cycle joining 13 orbits) give nearly or completely disjoint columns: coverage grows but total solver work is 137x-513x (Frobenius) and 5052x (horizontal) the best single chart.
- No natural public relation row was found, so target-RHS rank, sparse linear algebra, extraction and final DLP verification remain null. At m=83 the signed-Frobenius rho baseline is $2^{37.14}$ operations.

How to read the report

Exact statements are marked structural or exhaustive. Finite solver observations name k, curve, target selection and presentation. Censored means a CPU-time cap was reached and no degree is inferred. Observed F4 degree is never substituted for formal degree of regularity. No discrete logarithm was computed.

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ECC2K-130 and F_(2^83) side by side

The two studies run the same protocols on two members of one family, the Koblitz curve $y^2 + xy = x^3 + 1$. The structural differences below are what changes the engineering at $m=83$; the security conclusion does not change.

Quantity	ECC2K-130 (m=131)	F_(2^83) (m=83)
prime subgroup	129 bits	81 bits
[O_K : Z[pi]]	263 x 146505763881528721	6473 x 53676929
nearest conductor prime	263, split	6473, inert
floor / Frobenius orbits	262 = 2 x 131	6,474 = 78 x 83
horizontal I-isogenies from E0 at the conductor prime	2	0
pi on E0[l]	-l (torsion over F_(q^2))	2514 l (x-coordinates over F_(q^3236))
minimum noninteger endomorphism degree on the floor	121,046	73,324,526
large-subgroup embedding degree	118 bits	74 bits
floor constructed from	H_D (Arb), 262 roots	gamma_2 class polynomial (PARI CRT), 6474 roots
E0 Frobenius collision	2P_3 + P_5 + P_17 = O	2P_w + P_(w^2) - P_(w^4) = O
E0 duplicate excess k=5/6/7	8 / 28 / 60	20 / 44 / 88
Run-08 best ratio at k=16	0.9766	0.9772
signed-Frobenius rho	2^60.81	2^37.14
natural public relation rows	0	0

Evidence map and claim layers

Layer	Runs	Establishes	Does not establish
Structure	s00, inventory, 04, 05, 10 graph	orders, conductors, class numbers, explicit maps, torsion, adjacency	solver hardness or attack speed
Finite factor base	01, 02, 03, 08	exact rational membership, tags, relation images, selected regularity	cryptographic-scale natural yield
Presentation and solver	06, 07	direct S5, pair presentations, F4 traces, censored boundaries	asymptotic solving exponent
Multi-curve atlases	09, 10	column and target overlap, map paths, finite cost ratios	benefit from an untested shared solver
End-to-end	06-10	which gates remain null and why	a discrete logarithm

Reproducibility scale

- 6,475 curves: E0 and the complete conductor-6473 floor.
- 32,375 exact three-summand census rows ($k = 4..7$, three bases) and 19,425 $k=4$ solver cells.
- 1,243,200 Run-08 screening cells and 58,266 scaling cells.
- 38,844 certified horizontal edges over six split primes.
- A 268,563-bit ladder in $F_{(2^{268588})}$ for each of E0 and a floor curve; 6,474 degree-6473 kernel lines.

1. Parameters and the maximal-order crater

Quantity	Value	Consequence
Curve	$y^2 + xy = x^3 + 1$	defined over F_2 ; field $F_2[z]/(z^{83}+z^7+z^4+z^2+1)$
Field	$q = 2^{83}$	prime extension degree: no intermediate subfield
Group order	4 ELL, ELL = 2,417,851,639,230,796,216,685,689	cofactor four; ELL prime (Lucas certificate)
Trace over F_q	-6,151,469,093,347	fixes the isogeny class
Base Frobenius	$\tau^2 + \tau + 2 = 0$	norm-2 self-endomorphism
CM discriminant	-7	maximal order O_K
Embedding degree of ELL	14,565,371,320,667,447,088,468	74 bits; $(\text{ELL}-1)/k = 166$

$$\tau^{83} = -3249459927382 + -347450761417 \tau$$
$$[O_K : \mathbb{Z}[\pi]] = 347,450,761,417 = 6473 \times 53676929 \text{ (both primes inert in } \mathbb{Q}(\sqrt{-7}) \text{)}$$

All possible orders over $F_{(2^83)}$

Conductor	Discriminant	Class number	Min noninteger endomorphism degree
1	-7	1	2
6,473	-293,298,103	6,474	73,324,526
53,676,929	-20,168,488,948,097,287	53,676,930	5,042,122,237,024,322
347,450,761,417	-845,054,221,264,771,390,935,223	347,504,444,820	211,263,555,316,192,847,733,806

Largest structural discontinuity

At conductor 6473 the minimum noninteger endomorphism degree jumps from 2 to 73,324,526. The degree-2 Frobenius remains a relative map to the conjugate curve but is not a self-endomorphism of a strict descendant.

2. Volcano geometry and endomorphism orders

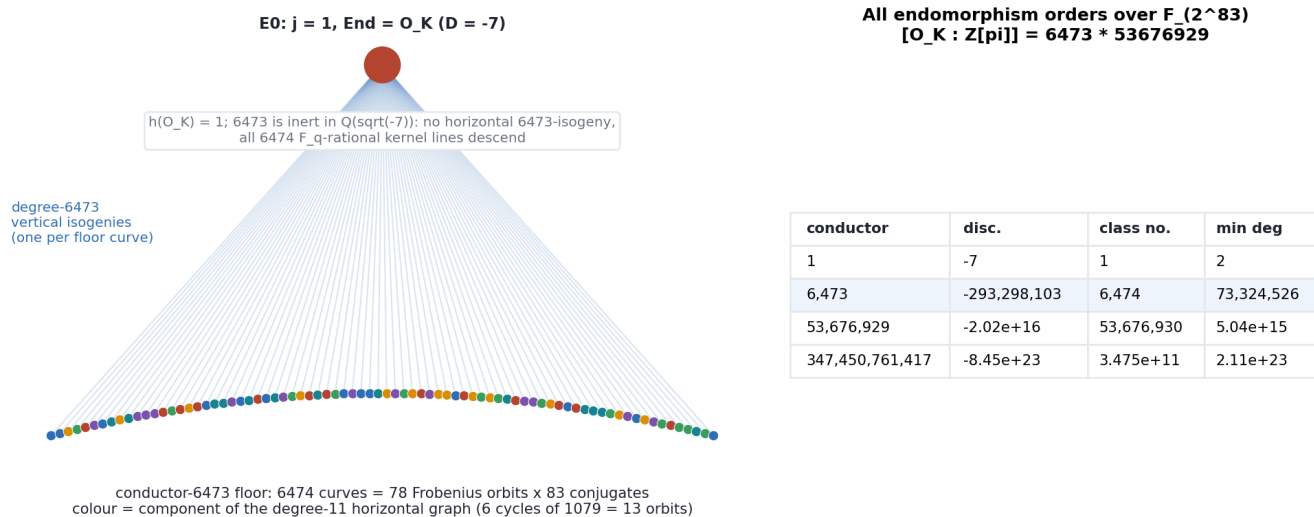


Figure 1. The degree-6473 volcano. E0 is the unique crater vertex ($h(O_K) = 1$); 6473 is inert, so all 6474 rational kernel lines descend, one to each floor curve. Floor vertices are aggregated into the 78 Frobenius orbits and coloured by their degree-11 horizontal cycle.

Property	E0 crater (f = 1)	Floor descendant (f = 6473)
Endomorphism order	O_K	Z + 6473 O_K
Discriminant	-7	-293,298,103
Class count	1	6,474
Degree-2 self-Frobenius	present	absent
Model over F_2	present	impossible (j has degree 83 over F_2)
Smallest noninteger norm	2	73,324,526
pi on E[6473]	2514 I	2514 I + N, N nilpotent
twist over F_(q^3236): 6473-part	(Z/6473)^2	Z/6473^2
Frobenius-stable 6473-kernel lines	6474	1
Full E[6473] field	F_(q^6472)	F_(q^41,893,256)
Prime subgroup, embedding degree	ELL, 74 bits	isomorphic, same 74 bits

3. The complete conductor-6473 floor

$D = -7 \times 6473^2 = -293,298,103$ has class number 6,474, so the direct Hilbert class polynomial would have coefficients of about 600,000 bits. The floor was therefore built from the $\gamma_2 = j^{(1/3)}$ class polynomial (PARI polclass, CRT method, 703 s; coefficients up to 202,143 bits). Cubing is a bijection on $F_{(2^{83})}^*$ because 3 does not divide $2^{83} - 1$, so every root r gives $j = r^3$.

- Modulo 2 the polynomial is squarefree and factors as 78 irreducibles of degree 83.
- Every one of the 6,474 curves $y^2 + xy = x^3 + b$, $b = 1/j$, has group order 4 ELL, certified by a point with $[4 \text{ ELL}]R = O$ and $[4]R \neq O$ (4 ELL is the only multiple of ELL in the Hasse interval); PARI SEA confirms one curve per orbit.
- $Cl(O_{6473})$ is cyclic of order 6,474; the class of a prime above 2 has order 83, matching the orbit length.
- Curves are named $O<\text{orbit}>-<\text{shift}>$: $j = j_0^{(2^{\text{shift}})}$ for the orbit's smallest-encoding j_0 .

What varies without changing the DLP group

Varies across the floor	Invariant
j-invariant and coefficient b	field q and trace
rational x density in a chosen subspace	group order 4 ELL
four-torsion tag distribution	prime subgroup order ELL
membership ANF degree and sparsity	embedding degree of ELL
target collisions in a finite factor base	abstract DLP scalar under a verified isogeny
Groebner presentation and solver trace	conductor 6473 and class number 6474
relative-Frobenius and horizontal charts	Frobenius orbit length 83

4. Run-01: all-curve finite comparison

Run-01: exact relation coverage and sampled regularity on E0 and all 6474 descendants

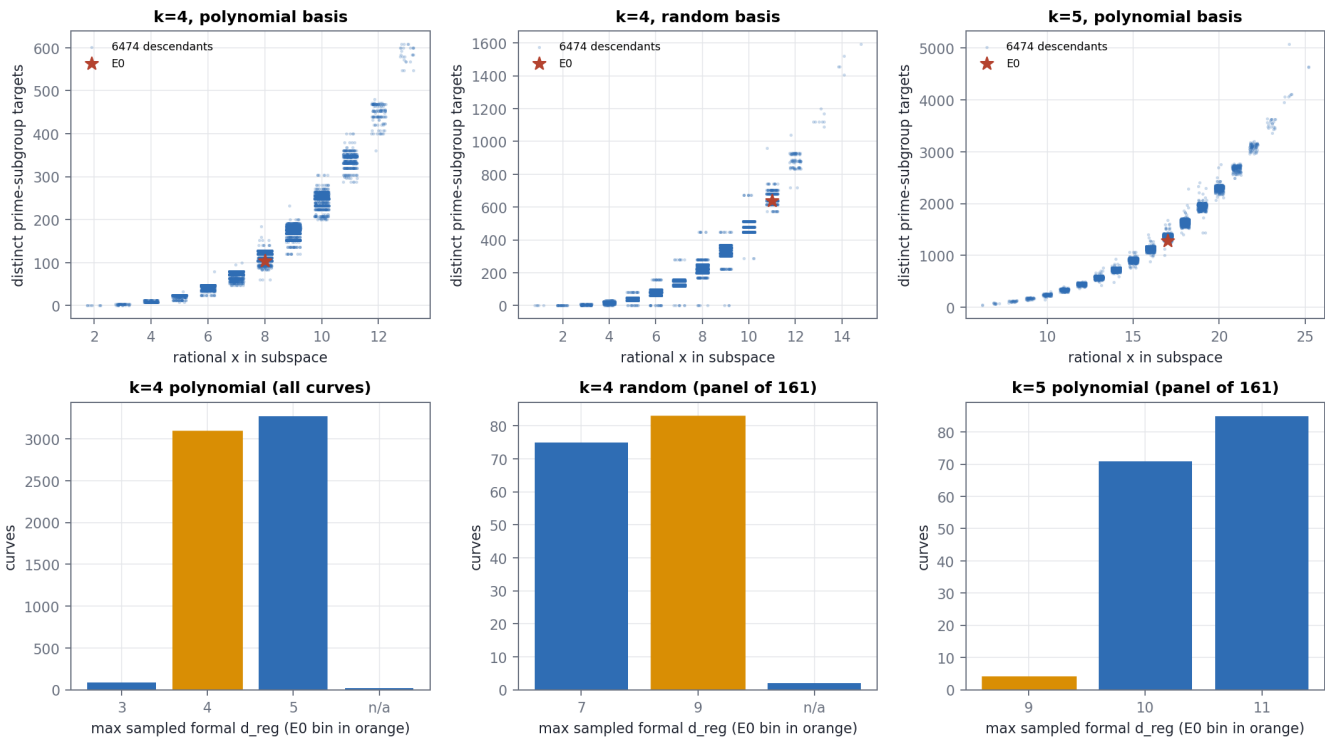


Figure 2. Top: exact distinct prime-subgroup targets against rational x count for all 6475 curves in the three Run-01 profiles (E0 starred). Bottom: maximum sampled formal regularity over two satisfiable targets: all curves at k=4 polynomial, and the 161-curve panel for the costlier profiles.

Profile	Floor x range	Coverage range	E0 x / coverage	Descendants above E0
4 polynomial	2-13	0-608	8 / 104	3,650
4 random	1-15	0-1,592	11 / 640	251
5 polynomial	6-25	32-5,072	17 / 1,280	2,827

Solver cells	Curves	Cells	Max sampled d_reg distribution	E0
k=4 polynomial	6,475	19,425	3: 85, 4: 3,094, 5: 3,276, None: 20	[4, 4]
k=4 random (panel)	161	483	7: 75, 9: 84, None: 2	[9, 9]
k=5 polynomial (panel)	161	483	9: 5, 10: 71, 11: 85	[9, 9]

Every completed cell was verified: the quotient dimension equals the ordered x-solutions of the exact group law, the input generators reduce to zero, and every group-witness assignment replays. Caps are CPU-time (20 s per stage) rather than ECC2K-130's wall-clock caps; no k=4 census cell was censored.

Run-01 interpretation

At k=4 polynomial E0 samples d_reg 4 with 3,276 descendants above it and 85 below. The panel profiles are stratified by the membership indicator: k=4 random gives 7 for an even rational-x count and 9 for an odd one, and at k=5 the maximum sampled d_reg equals 6 plus the membership-ANF degree for 160 of 160 panel descendants. E0 is the one k=5 curve below its stratum (9 instead of 11); section 6 attributes this.

5. Factor-base construction and exact yield accounting

For a k -dimensional F_2 -subspace W , the factor base keeps nonzero x in W that lift to rational points: on $y^2 + xy = x^3 + b$ this is $\text{Tr}(x + b/x^2) = 0$. Each x has two signed lifts; $[\text{ELL}]P$ lies in the cyclic $E[4]$ and gives a $Z/4$ tag. A relation for a prime-subgroup target needs total tag 0.

$$\text{eligible} = 8 \left(C(n_0, 3) + n_0 C(n_2, 2) \right) + 4 \left(n_0 + n_2 \right) C(n_0, 2)$$

The formula matched 32,375 of 32,375 census rows ($k = 4, 5, 6, 7$; polynomial and random bases). No row contained an infinity sum, and every row except E0 at $k = 5, 6, 7$ was injective.

Profile	Floor x range	Coverage range	E0 x	E0 eligible / distinct
4 polynomial	2-13	0-608	8	104 / 104
4 random	1-15	0-1,592	11	640 / 640
5 polynomial	6-25	32-5,072	17	1,300 / 1,280
6 polynomial	16-46	1,096-30,448	39	18,116 / 18,072
7 polynomial	40-82	19,760-177,632	68	100,120 / 100,032

6. Run-02: pinning down the differences

Frobenius collision identity

At $m=83$ the collision chain is $x = w, w^2, w^4$ (masks 2, 4, 16): $x = 1 + w$ does not lift on E0 here, so ECC2K-130's chain $1+t, 1+t^2, 1+t^4$ is replaced. With canonical signs $\tau(P_w) = +P_w(w^2)$ and $\tau^2(P_w) = -P_w(w^4)$, and $\tau^2 + \tau + 2 = 0$ becomes

$$2 P_w + P_w(w^2) - P_w(w^4) = 0$$

Row	Eligible	Distinct	Excess	Multiplicity	Common masks	Difference class
E0 k=5 polynomial	1,300	1,280	20	2 (all)	5	$+(2e_w + e_{w^2} - e_{w^4}) (20/20)$
E0 k=6 polynomial	18,116	18,072	44	2 (all)	11	$+(2e_w + e_{w^2} - e_{w^4}) (44/44)$
E0 k=7 polynomial	100,120	100,032	88	2 (all)	22	$+(2e_w + e_{w^2} - e_{w^4}) (88/88)$

Across all 32,375 census rows the only non-injective images are these three E0 rows; each excess is exactly four duplicate targets per compatible common mask.

Exhaustive target correction (k=4 polynomial)

Curve	Selection	Target x	Row-reduced d_reg distribution	Raw top rank / d_reg
E0	original	52	d=4: 40, d=5: 8, d=6: 2, d=7: 2	36/10
O30-39	same rational-x count as E0, maximum coverage	92	d=4: 92	36/10
O26-20	same rational-x count as E0, minimum nonzero coverage	30	d=4: 30	36/10
O64-62	largest k=4 polynomial coverage	304	d=5: 304	36/10
O53-10	sampled dreg-3 representative with the most targets	152	d=3: 152	36/7
O00-00	orbit O00 representative	28	d=5: 28	36/10

Every descendant has a single row-reduced degree across all of its reachable targets (5 of 5 selected descendants are uniform). E0 is the only target-dependent curve: 12 of its 52 target-x systems exceed its typical degree 4. None of E0's exceptional targets is a Frobenius image of another target, and the raw highest-degree row space (rank 36, raw d_reg 10) is shared by every curve with a degree-4 membership indicator. As at ECC2K-130, a sampled maximum on E0 can land on an exceptional target; it is not evidence that descendants are generically easier.

One-axis counterfactuals (E0 anchor, k=4 polynomial)

Axis	Systems	Raw top row spaces	Raw d_reg	Row-reduced d_reg	Unit ideal
coefficient	6,475	1	10: 6475	0: 6448, 4: 27	6,448
membership	1,023	32	7: 16, 8: 16, 10: 991	3: 14, 4: 498, 5: 511	0
target	1,000	1	10: 1000	0: 999, 4: 1	999

The coefficient axis sweeps all 6475 curve coefficients, the membership axis every distinct k=4 membership pattern and the target axis a deterministic 1000-target sample of the census targets; the counts above are the systems completed when this report was built. Every completed coefficient and target substitution has the same raw highest-degree row space (rank 36) and raw d_reg 10, exactly as at ECC2K-130; only the membership axis changes it. Counterfactual systems need not be natural factor bases for the substituted curve; an inconsistent (unit) system is a warning, not an easier system, and is excluded from structural comparisons.

Why E0 is below its stratum at k=5

System family	Systems	Raw d_reg	Row-reduced d_reg
E0 anchor (b = 1, E0 membership, E0 target)	1	13: 1	9: 1
floor coefficient b, E0 membership and target	24	13: 24	11: 24
b = 1, descendant degree-5 membership, E0 target	24	13: 24	9: 24

Replacing only E0's coefficient $b = 1$ by floor coefficients restores the degree-5 membership stratum (24 of 24 systems reach 11); with E0's b and 24 descendant degree-5 membership patterns, 24 systems stay at 9. The effect therefore travels with the coefficient $b = 1$, which lies in F_2 ; a plausible mechanism is that the S_4 coefficient structure over $F_{(2^{83})}$ degenerates before row reduction, but only the attribution is measured here. Only the crater can have a coefficient in F_2 (a strict descendant has j of degree 83), so this is a property of E0 and not a descendant advantage; in this presentation it makes E0, not the descendants, the lower-degree system.

7. Run-03: scaling, collisions and coefficient rank

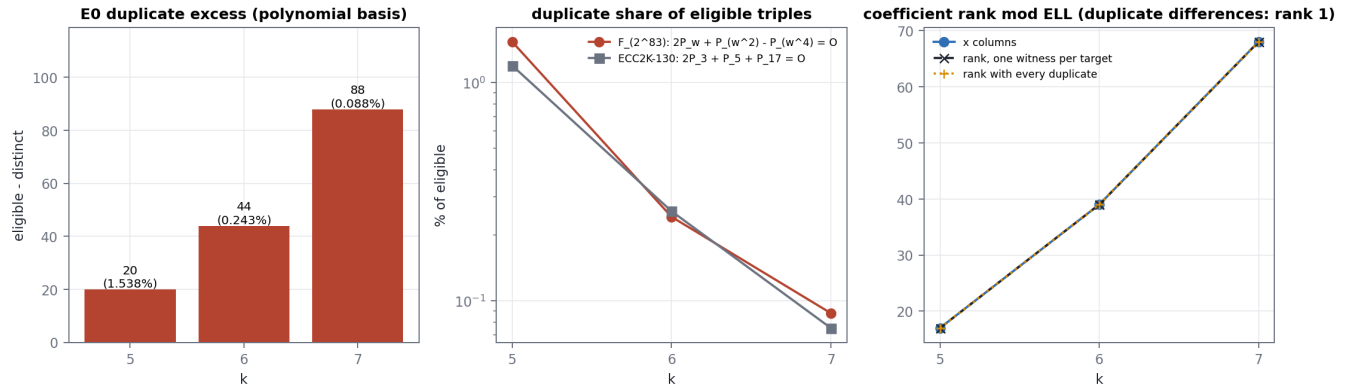


Figure 3. E0's Frobenius duplicate excess grows in count but shrinks as a share of eligible triples, tracking ECC2K-130; duplicates never add factor-base rank.

k	x columns	Distinct targets	Duplicate excess	Rank (one witness/target)	Rank with duplicates	Duplicate-difference rank
5	17	1,280	20	17	17	1 (identity: yes)
6	39	18,072	44	39	39	1 (identity: yes)
7	68	100,032	88	68	68	1 (identity: yes)

The all-curve census extends ECC2K-130's k=6 census to k=7: floor x counts span 16-46 (k=6) and 40-82 (k=7), coverage 1,096-30,448 and 19,760-177,632, and E0 is the only non-injective curve in both.

Selected regularity cells (one target per curve, 30 s CPU cap)

Curve	k=5	k=6	k=7
E0	10 (raw 11)	censored	censored
O26-20	10 (raw 12)	censored	censored
O53-10	11 (raw 13)	censored	censored
O30-39	10 (raw 12)	censored	censored
O64-62	11 (raw 13)	censored	censored
O00-00	11 (raw 13)	censored	censored

8. Groebner metrics: what was measured

Metric	Definition here	Interpretation
Formal degree of regularity	1 + highest standard-monomial degree of the ideal of top homogeneous components	presentation diagnostic
Observed max F4 degree	largest degree in msolve F4 rounds	solver trace; not formal d_reg
Quotient dimension	standard monomials of the affine basis	ordered Boolean solutions (verified)
CPU cap	process CPU seconds per stage (msolve: RLIMIT_CPU)	censored = null, never a degree
Wall and CPU time	recorded separately for every stage	implementation and load dependent

For every normalized binary curve $y^2 + xy = x^3 + a x^2 + b$ the third summation polynomial is

$$S_3(x_1, x_2, x_3) = (x_1 x_2 + x_1 x_3 + x_2 x_3)^2 + x_1 x_2 x_3 + b$$

The coefficient a disappears and b is a constant term: descent does not lower the formal degree of S_3 or remove monomials. Measured differences arise after coordinate substitution, membership, row reduction and target choice.

9. Explicit vertical descent and public-point transport

The ECC2K-130 route (Velu formulas on $E_0[263]$ over $F_{(q^2)}$) is impossible here: 2514 is a primitive root mod 6473, so kernel x-coordinates first appear over $F_{(q^{3236})} = F_{(2^{268588})}$. They are points of the quadratic twist of E_0 over that field, whose 6473-part is $(Z/6473)^2$.

Method

- Tower arithmetic $F_{(2^{83})}[Y]/(Y^{3236} + Y^{887} + 1)$: Kronecker packing into 165-bit slots, PMULL Karatsuba, Itoh-Tsujii inversion; trace to F_q is a sum of selected coefficients.
- One x-only Montgomery ladder by the 268,563-bit twist cofactor gives $x(P)$ of order 6473 (7005 s).
- $\tau(P + \tau P) = -2P$ gives $x(P + \tau P) = x_P + 1/x_P$, so the lines $\langle P + v \tau P \rangle$ and $\langle \tau P \rangle$ are walked by differential additions without any y-coordinate or Artin-Schreier solve.
- Frobenius permutes a line's 3236 +-classes transitively, so the Velu sum is $v = \text{Tr}_{(F_Q/F_q)} x(G)$ and the codomain is $y^2 + xy = x^3 + b'$ with $b' = 1 + v + v^2$.
- Kernel polynomials (degree 3236) come from Berlekamp-Massey on $\text{Tr}(x^i)$; since E_0 is defined over F_2 , a conjugate line has the squared kernel polynomial, so only the cheapest line per orbit is computed.
- Every step was first validated against Sage on $m = 17$, $l = 271$ (torsion field $F_{(2^{2295})}$): 272/272 codomains equal the class-polynomial roots and the kernel polynomials divide the 271-division polynomial.

Result	Value
pi on the torsion point	$x(P)^q = x([2514]P)$: verified
kernel lines	6,474
horizontal maps back to $j = 1$	0
distinct codomains	6,474
inventory curves matched	6,474 / 6,474
chain closes ($G_{6473} = P$)	yes
line enumeration (native)	837 s

Selected explicit maps

Descendant	Rule	Line	Computed from	Kernel degree	Kernel SHA-256
O14-67	E_0 x-count at $k=5$, maximum $k=5$ coverage	4992	O14-01 (14 steps, σ^{66})	3236	f8a85a57af055578...
O76-25	E_0 x-count at $k=5$ and $k=6$, maximum $k=5$ coverage	3221	O76-39 (74 steps, σ^{69})	3236	6bbfc51973293065...
O51-49	E_0 x-count at $k=4, 5$ and 6 , maximum $k=5$ coverage	2689	O51-47 (70 steps, σ^2)	3236	beface526f300d47...

Sage's generic isogeny constructor stalls on a degree-3236 kernel polynomial (it runs an NTL irreducibility test over $F_{(2^{83})}$), so the maps are evaluated in closed form. In characteristic 2, summing $x(P+Q) + x(P-Q) = x_P x_Q / (x_P + x_Q)^2$ over the kernel and using $\psi'' = 0$ gives

$$x(\phi(P)) = x + u^2 + u, \quad u = x \psi'(x) / \psi(x)$$

onto $y^2 + xy = x^3 + v x + (1 + v)$, v the sum of the roots of ψ , normalized by $y \rightarrow y + v$ to $b' = 1 + v + v^2$. The formula was checked against Sage's isogeny on the $m=17$ toy (60 points, 3 lines). For every map the codomain equals the inventory curve; the images of the header's P and Q lie on it, are nonzero and are killed by ELL; x-only commutation holds for 2, 3 and 6473; additivity holds on random point pairs; and $\gcd(6473, \text{ELL}) = 1$ makes the map injective on the prime subgroup. The sign of $\phi(Q)$ is fixed by $x(\phi(P) + \phi(Q)) = x(\phi(P + Q))$ (a global sign change is again an isogeny). No logarithm was computed.

Descendant	Native kernel	Codomain setup	One point map
O14-67	202 s	1.78 s	0.436 s
O76-25	182 s	1.15 s	0.448 s
O51-49	163 s	1.45 s	0.613 s

10. Embedding degree and conductor-prime torsion

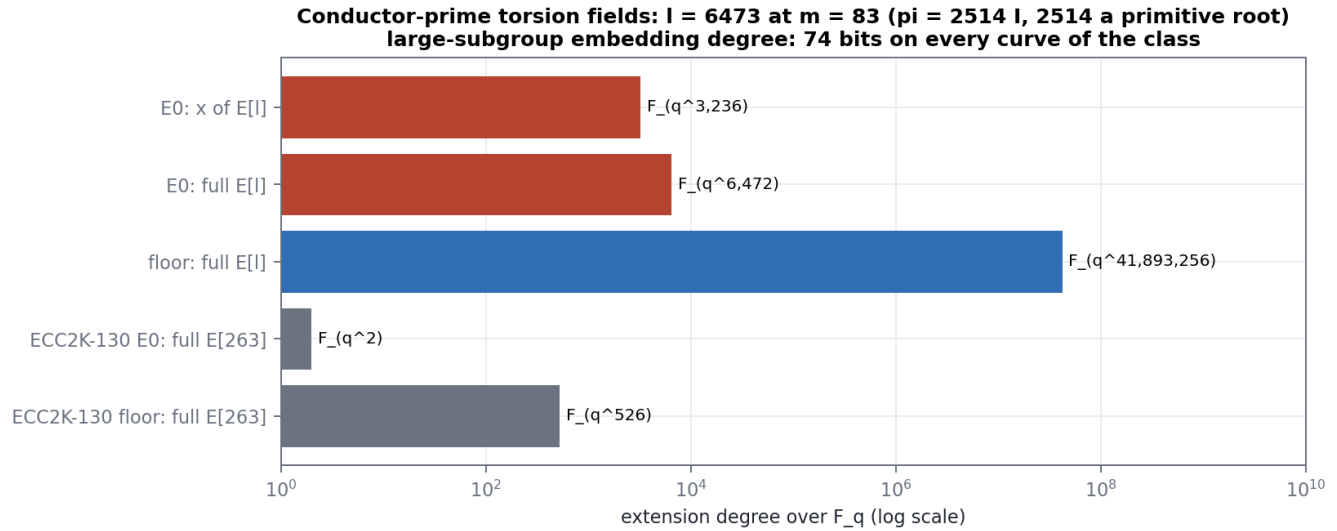


Figure 4. Torsion fields at the conductor prime. At $m=83$ descent moves full $E[6473]$ from $F_{(q^{6472})}$ to $F_{(q^{41893256})}$; at ECC2K-130 it moved $E[263]$ from $F_{(q^2)}$ to $F_{(q^{526})}$.

Quantity	E0	Conductor-6473 descendant
ord_ELL(q)	14,565,371,320,667,447,088,468	same
embedding-degree bits	74	same
$q \bmod 6473$, $t \bmod 6473$	2548, 5028	same
Frobenius polynomial mod 6473	$(X - 2514)^2$	same
μ_{6473} embedding degree	3236	same
π on $E[6473]$	$2514 * l$	$2514 * l + N$, N nilpotent nonzero
twist over $F_{(q^{3236})}$: 6473-part	$(Z/6473)^2$	$Z/6473^2$ (point of order 6473^2 found on O00-00)
stable kernel lines	6,474	1
full $E[6473]$ field	$F_{(q^{6472})}$	$F_{(q^{41893256})}$

Security implication

The prime-subgroup embedding degree is a 74-bit integer, invariant on the isogeny class: descent gives no pairing reduction for ELL. The 6473-torsion difference is useful for navigation and kernel validation only, and it is far more expensive to reach than ECC2K-130's 263-torsion.

11. Run-06: matched four-summand end-to-end gates

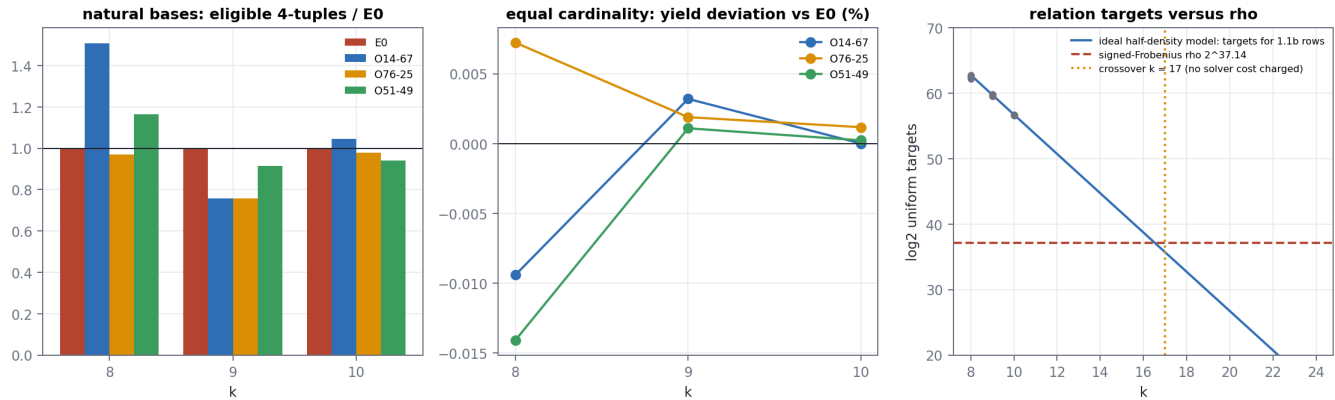


Figure 5. Natural-base raw yield, equal-cardinality deviation and relation targets against the rho baseline.

k	Curve	x count	Eligible 4-tuples	Raw yield vs E0	Attempt reduction vs E0
8	E0	131	46,876,448	1.0000x	+0.00%
8	O14-67	145	70,669,344	1.5076x	+26.81%
8	O76-25	130	45,444,960	0.9695x	-1.73%
8	O51-49	136	54,534,440	1.1634x	+11.08%
9	E0	270	866,203,128	1.0000x	+0.00%
9	O14-67	252	656,272,032	0.7576x	-23.54%
9	O76-25	252	656,263,448	0.7576x	-23.55%
9	O51-49	264	791,326,080	0.9136x	-7.25%
10	E0	524	12,421,907,088	1.0000x	+0.00%
10	O14-67	530	13,002,424,200	1.0467x	+3.47%
10	O76-25	521	12,139,210,440	0.9772x	-1.80%
10	O51-49	516	11,678,473,296	0.9402x	-4.71%

Equal-cardinality and common-x controls

k	Matched x	Deviations vs E0	Common x	Common-x deviations
8	130	O14-67 -0.0094%, O76-25 +0.0072%, O51-49 -0.0141%	21	O14-67 -1.066%, O76-25 +0.000%, O51-49 -0.733%
9	252	O14-67 +0.0032%, O76-25 +0.0019%, O51-49 +0.0011%	34	O14-67 -0.344%, O76-25 +0.000%, O51-49 +0.069%
10	516	O14-67 +0.0000%, O76-25 +0.0012%, O51-49 +0.0002%	67	O14-67 -0.058%, O76-25 +0.009%, O51-49 -0.091%

Natural public-target probes (k = 10 pair tables)

Curve	Pair entries	Distinct sums	Pair collision excess	Public targets	Natural rows	Planted recovered
E0	548,104	548,094	10	4	0	2/2
O14-67	560,740	560,740	0	4	0	2/2
O76-25	541,840	541,840	0	4	0	2/2
O51-49	531,480	531,480	0	4	0	2/2

Public targets are $Q' + [a_i]P'$ with $a_i = \text{SHA256}(\text{"m83-run06-natural-target:"} i) \bmod \text{ELL}$ on the Run-04 images of the header points. The collision-free coverage bound per uniform target is about $2^{47.4}$, so zero natural rows is the expected outcome; planted controls validate lookup and group replay only.

Direct S5 versus enumerative pair control (k=4, planted targets)

Curve	x	Direct vars	Direct formal d_reg	Direct max F4	Pair vars	Pair formal d_reg	Pair max F4
E0	8	16	censored	12	10	8	10
O14-67	11	16	censored	12	12	10	12
O76-25	7	16	censored	12	10	8	10
O51-49	8	16	censored	12	10	8	10

As at ECC2K-130, the pair presentation lowers the observed F4 degree from 12 to 10 only on the small 7-8 x bases and not on the 11-x base of O14-67: the effect follows factor-base size, not conductor. The pair index is an enumerative finite control; its table is quadratic in the factor base.

Direct S5 versus enumerative pair control (k=5, planted targets)

Curve	x	Direct vars	Direct formal d_reg	Direct max F4	Pair vars	Pair formal d_reg	Pair max F4
E0	17	20	censored	censored	16	censored	censored
O14-67	17	20	censored	censored	16	censored	censored
O76-25	17	20	censored	censored	16	censored	censored
O51-49	17	20	censored	censored	16	censored	censored

Rho and useful-scale comparison

Baseline	log2 group operations
no quotient	40.826
negation quotient	40.326
signed Frobenius quotient (166)	37.138
relation targets for 1.1b rows, k=8 (best curve)	62.25
relation targets for 1.1b rows, k=9 (best curve)	59.52
relation targets for 1.1b rows, k=10 (best curve)	56.59

Under an ideal half-density, uniform-tag, collision-free extrapolation the relation-target count crosses the rho line at $k = 17$. That point still charges no S5 solve (68 Boolean root variables), needs about 2^{16} columns and a 2^{31} -entry pair table, sparse linear algebra, extraction and verification.

12. Run-07: exact half-trace pair membership

k	Curve	Pair vars	Pair ANF degree	Pair ANF terms	Density	Direct degree	Direct terms
4	E0	11	10	809	0.395	3	9
4	O14-67	11	11	807	0.394	4	9
4	O76-25	11	11	753	0.368	4	9
4	O51-49	11	10	533	0.260	3	4
5	E0	14	13	7,551	0.461	5	19
5	O14-67	14	13	6,777	0.414	5	18
5	O76-25	14	13	6,893	0.421	5	16
5	O51-49	14	13	6,257	0.382	5	17
6	E0	17	17	63,143	0.482	6	37
6	O14-67	17	17	60,523	0.462	5	40
6	O76-25	17	17	60,595	0.462	6	30
6	O51-49	17	17	58,717	0.448	6	32
7	E0	20	19	514,327	0.491	6	68
7	O14-67	20	20	501,949	0.479	6	63
7	O76-25	20	19	490,721	0.468	7	65
7	O51-49	20	19	485,949	0.463	7	64

Every exhaustive (s, t) evaluation accepted exactly $C(b, 2)$ pairs (64 half-trace identity checks passed). At the Run-06 crossover $k = 17$: direct S5 uses 68 Boolean variables, two affine pair images 100, a root witness 68, a full-field auxiliary circuit 598; pair indexing needs about 2^{31} pairs and a 2^{62} -cell two-index table to save two semantic bits.

13. Run-08: complete-floor adapted subspaces

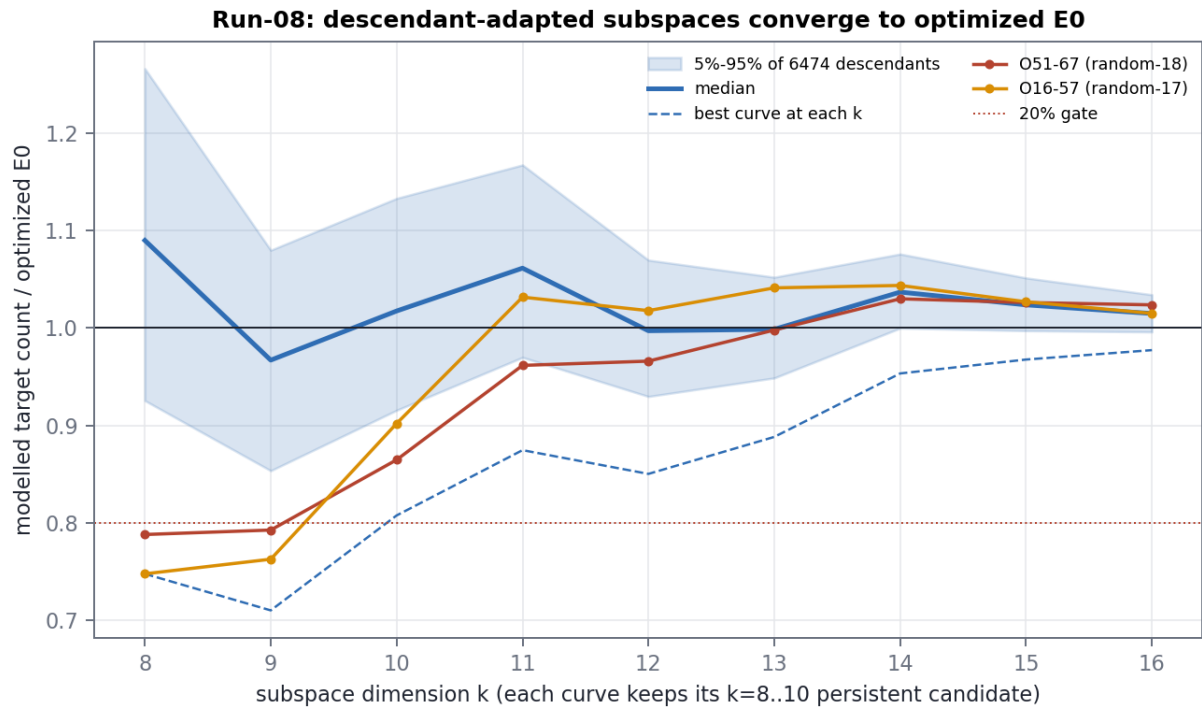


Figure 7. All 6474 descendants, each with its k=8..10 persistent candidate among 64, scaled to k=16 against optimized E0. The two curves that clear the gate at k=8/9 are traced.

k	Best	Best curve	Median	5%-95%
8	0.7477	O16-57	1.0897	0.9254-1.2659
9	0.7102	O52-53	0.9669	0.8535-1.0797
10	0.8076	O43-39	1.0173	0.9155-1.1328
11	0.8746	O51-33	1.0613	0.9698-1.1671
12	0.8502	O21-53	0.9970	0.9296-1.0695
13	0.8882	O49-54	0.9985	0.9486-1.0519
14	0.9532	O25-28	1.0367	0.9997-1.0757
15	0.9675	O59-01	1.0236	0.9971-1.0512
16	0.9772	O42-74	1.0147	0.9958-1.0340

Gate passer	Candidate	Exact ratio k=8	Exact ratio k=9	Ratio k=10	Ratio k=16
O17-77	scale-30	0.7623	0.7927	0.9573	0.9980
O51-67	random-18	0.7881	0.7926	0.8647	1.0236

The screen evaluated 1,243,200 cells; optimized E0 uses random-11. Unlike ECC2K-130, where no finalist passed, two of 6474 curves clear the 20% adjacent-dimension gate with no-worse membership degree, both only at k=8/9. Neither persists: no curve passes at any adjacent pair from k=9/10 on, and the finalists' exact tag correction is at most 0.029%. The early leads are finite density fluctuations selected from 6474 curves.

14. Run-09: Frobenius-conjugate atlases

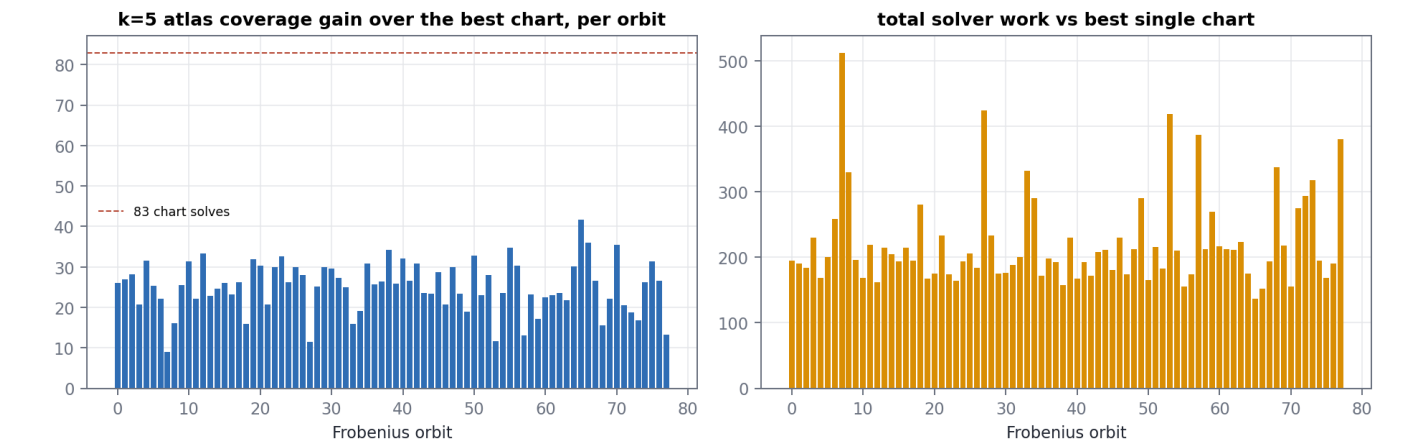


Figure 8. Per-orbit coverage gain and total solver work of the 83-chart Frobenius atlas at k=5, for all 78 orbits.

Measurement	All 78 orbits
k=10 pulled column union per orbit	44,661-45,123
union / largest chart (k=10)	76.9x-80.1x
k=10 column overlap excess, all orbits	73
k=5 cross-chart target overlap, all orbits	8
k=5 coverage gain vs best chart	8.9x-41.7x
total solver work vs best chart	137x-513x
common transported basis: identical columns and targets	all orbits

Sixteen public targets $Q' + [a_i]P'$ on each mapped descendant, transported by relative Frobenius to all 83 charts of its orbit and matched against each chart's $k=9$ pair table, produced 0 natural rows over 3,984 chart-target cells.

Chart-specific bases give new coverage but almost no shared columns: across all 78 orbits only 73 of the $k=10$ pulled columns and 8 of the $k=5$ targets are shared between charts. A common transported basis gives identical columns and targets on every chart: one-fold coverage for 83-fold work. ECC2K-130's two orbits gave 297x-317x total work; at $m=83$ the range over 78 orbits is 137x-513x.

15. Run-10: separable horizontal-isogeny atlas

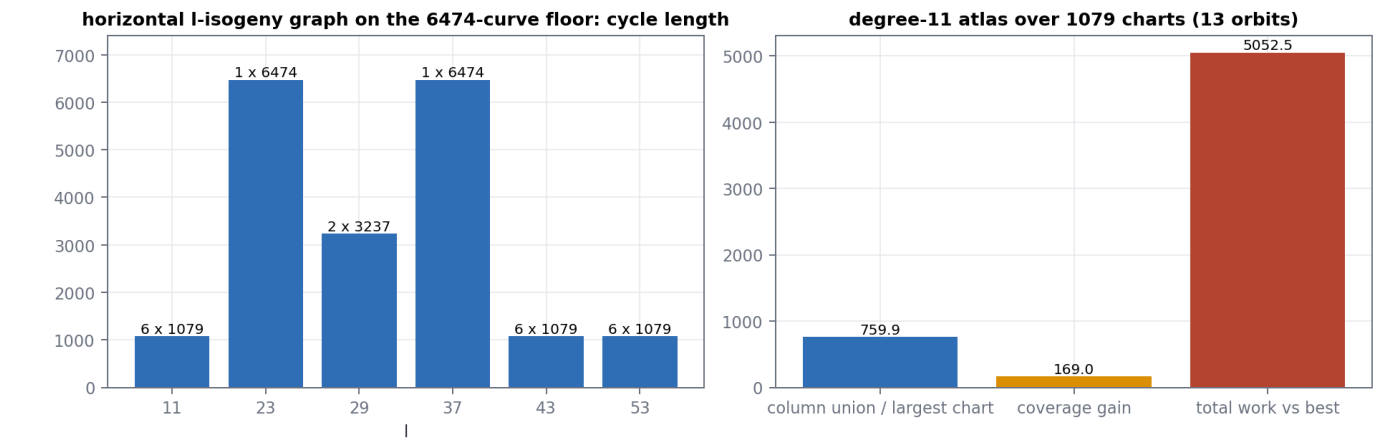


Figure 9. Left: cycle structure of the horizontal I-isogeny graphs on the full floor. Right: the degree-11 atlas of the reference descendant's cycle.

l	Components	Cycle length	Class order	Orbits per cycle
11	6	1,079	1,079	13
23	1	6,474	6,474	78
29	2	3,237	3,237	39
37	1	6,474	6,474	78
43	6	1,079	1,079	13
53	6	1,079	1,079	13

Φ_l (PARI polmodular, reduced mod 2 and hash-pinned) was specialised exactly at all 78 orbit representatives and at 156 further vertices; Frobenius equivariance of Φ_l over F_2 gives every other edge. Each graph is 2-regular and its cycle length equals the order of the class of a prime above l in the cyclic group $Cl(O_{6473})$. Unlike ECC2K-130, where each 11-cycle was one Frobenius orbit, an 11-cycle here joins 13 orbits ($1079 = 13 \times 83$).

Measurement	Degree-11 atlas of O14-67
charts (orbits)	1,079 (13)
average / maximum path length	269.7 / 539 edges
x-map evaluations	5,129,173
column union / overlap excess	18,998 / 0
atlas targets / cross-chart overlap	14,256,952 / 0
coverage gain vs best chart	169.0x
total solver work vs best chart	5052x

The 26 seed maps (two per orbit, Sage isogenies_prime_degree, a few seconds each) and their Frobenius conjugates give every edge of the cycle. Horizontal transport produces completely disjoint columns and targets, as Frobenius transport did, while each chart is now up to 539 degree-11 steps from the reference. Because the cycle holds 1079 charts, the atlas penalty grows to 5052x the best single chart (ECC2K-130: 297x-318x over 131 charts).

16. End-to-end mechanism comparison

Mechanism	Finite benefit	Complete cost found	Disposition
Static floor curve	up to 29% in single small-k cells	decays to about 1.0 by k=13-16	screening only
Equal-cardinality descendant	at most +0.0141%	no persistent yield difference	no attack gain
E0 Frobenius duplicates	lower k=5 regularity on E0	rank-one redundancy, no new rows	engineering cleanup
Half-trace pair membership	exact table-free evaluator	dense near-full-degree ANF or more variables	rejected representation
Common Frobenius atlas	shared columns	identical relations, 83x work	redundant
Chart-specific Frobenius atlas	coverage gain 9x-42x	nearly disjoint columns; 137x-513x total work	rejected
Degree-11 horizontal atlas	coverage gain 169x over 1079 charts	disjoint columns; 5052x total work plus 5,129,173 map evaluations	rejected
Pairing / embedding	local 6473 navigation	ELL embedding degree unchanged (74 bits)	no MOV shortcut

Gate status

Gate	Status	Evidence
explicit descendant maps	complete	all 6474 degree-6473 lines; selected kernel polynomials and P, Q transport
matched finite factor bases	complete	all 6475 curves; adapted subspaces to k=16
natural useful-scale relation yield	open	small bases give zero public rows; useful k not solved
complete Groebner/SAT cost	partial	k=4 complete; k=5 panel complete under CPU caps; larger k censored or structurally blocked
target-RHS rank	null	no natural public rows
sparse linear algebra, extraction, verification	null	no natural matrix
rho comparison	bounded	finite lower bounds against $2^{37.14}$ signed-Frobenius rho

17. Security interpretation

What is established

Volcano descent at $m=83$ changes the endomorphism ring and the 6473-torsion far more drastically than at ECC2K-130 (an inert conductor prime, a primitive-root Frobenius, torsion over $F_{(2^{268588})}$), and it multiplies the number of coordinate presentations by 25. Across the tested direct, pair, adapted-subspace, Frobenius-atlas and horizontal-atlas mechanisms, every apparent advantage is absorbed by factor-base size, membership complexity, matrix columns, solver multiplicity or map cost.

What remains unknown

No experiment collected cryptographic-scale natural four-summand relations or built a target-RHS matrix. A mechanism that changes decomposition-solver complexity, or creates reusable columns with independently new relations, could still matter. At $m=83$ the rho baseline ($2^{37.1}$) is small enough that any index calculus would have to beat it at k near 17 with no solver cost at all; the data give no reason to expect that.

Operational guardrails

- Keep natural public-target rows separate from planted controls.
- Charge factor-base construction, membership, equation assembly, solving, matrix columns, rank, linear algebra, extraction, verification and isogeny transport.
- Require adjacent-dimension persistence and at least 20% complete-cost improvement before scaling a lead.
- Preserve caps as censored data; never convert a timeout into a degree or hardness claim.

Appendix A. Protocols and sample sizes

Run	Scope	Scale	Boundary
s00	invariants	Lucas certificates, four orders	no hardness
inventory	gamma_2 class polynomial	78 x 83 curves, order certificates	none
01	all curves, three profiles	19,425 solver cells + 161-curve panel	finite S4 presentation
01/03 census	exact images k=4..7	32,375 rows	three summands
02	attribution, exhaustion, counterfactuals	9,205 systems	membership drives regularity
04	explicit degree-6473 maps	6474 lines, selected kernels	no DLP
05	embedding and 6473-torsion	two 268,563-bit ladders	local torsion only
06	four-summand gates	k=8..10 panels, public probes	zero natural rows
07	half-trace membership	exhaustive (s,t) through k=7	representation falsifier
08	adapted subspaces	1,243,200 + 58,266 cells	bounded candidate family
09	Frobenius atlases	78 orbits x 83 charts	no shared-cost solver
10	horizontal graph and atlas	38,844 edges; 1079-chart atlas	degree-11 paths charged

Censoring policy: a capped stage records the completed stages and a null observable; it is never pooled with completed degrees. Target policy: planted targets validate equations, root recovery and group replay; natural targets Q + [a]P test relation collection; planted rows never enter rank.

Appendix B. Audit, artifacts and references

`scripts/audit.py` re-derives the headline claims from the saved receipts, using Sage point arithmetic rather than the native library where possible, and writes `outputs/SHA256SUMS` over 150 files: 19 of 19 checks passed.

Check	Result
s00 invariants reproduce	pass
inventory: 78 orbits x 83 = 6474 distinct j	pass
class polynomial mod 2 hash	pass
sample floor curve order 4 ELL (Sage)	pass
6474 kernel lines, distinct codomains equal to the inventory	pass
Velu $j = 1/(1+v+v^2)$ recomputed for 200 lines	pass
map O14-67: images on the target, nonzero, killed by ELL; kernel hash; b matches inventory	pass
map O51-49: images on the target, nonzero, killed by ELL; kernel hash; b matches inventory	pass
map O76-25: images on the target, nonzero, killed by ELL; kernel hash; b matches inventory	pass
census rows = 6475 x 5	pass
eligibility formula on every census row	pass
only E0 at k=5,6,7 is non-injective	pass
census row O52-04 k=5 reproduced in Sage	pass
k=4 solver census: 6475 curves, no errors	pass
E0 twist point of order 6473; floor twist point of order 6473^2 ; $\pi = 2514$	pass
horizontal graphs 2-regular with cycle length = class order	pass
Run-06: zero natural rows, all planted controls recovered	pass
Run-09: zero natural rows	pass
Run-08: no gate passer at $k \geq 11$; screen covers 6475 x 64 x 3 cells	pass

Artifact	Location
invariants	outputs/s00-ring-invariants.json
floor inventory	outputs/inventory/inventory.json (class polynomial mod 2, SHA-256 of the integer file)
Run-01/03 census and solver cells	outputs/run01-comparison/
Run-02 attribution	outputs/run02-attribution/
Run-04 torsion, lines, maps	outputs/run04-explicit-descent/
Run-05 .. Run-10	outputs/run05-embedding.json, run06-four-summand/ .. run10-horizontal/
native code	native/ (tower field, isogeny tool, F_(2^83) curve library, CPU caps)
figures	outputs/figures/

Source	Use
ECC2K-130 volcano-descendant report, runs 01-10	protocols reproduced here; comparison column
Semaev, summation polynomials	S3/S4/S5 systems
Gaudry; Diem	extension-field index calculus context
Enge-Sutherland; PARI polclass	class invariants and CRT class polynomials
Sutherland; PARI polmodular	classical modular polynomials Φ_l
Lopez-Dahab	x-only binary-curve ladder
Kohel; Velu	isogenies from kernel polynomials
Wiener-Zuccherato; Gallant-Lambert-Vanstone	Frobenius-aware rho baselines